



Submitted By:

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Assignment:

Pacman Game

Submitted To:

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For the fulfillment of,

Object-Oriented Programming Lab

Department of Computer Engineering

University of Engineering and Technology, Lahore

Graphical Representation

PACMAN PROJECT — ARCHITECTURE

Model-View-Controller Design Pattern



Architecture Pattern: Model-View-Controller (MVC)

■ Model Layer ■ Controller Layer ■ View Layer ■ Input Handler

Project Report: Simple Java Pacman

1. Introduction

This project is a basic version of the classic Pacman game. It runs in the console and was built to practice Java and Object-Oriented Programming (OOP). It shows how classes and objects work together to create a simple game structure.

2. The Different Classes

The code is divided into five main classes to keep it organized:

- **GameBoard**
This class creates the 10x10 grid and handles drawing the board on the screen.
- **Pacman**
This class stores the player's position and controls movement in four directions: up, down, left, and right.
- **Ghost**
This class represents the enemy. It moves randomly across the grid.
- **Food**
This class represents the items Pacman needs to collect in order to win the game.
- **Game**
This is the main class that runs the game loop and manages win/loss conditions.

3. How the Game Works

The game runs inside a continuous loop (while loop) until the game ends. Each turn follows these steps:

- **Reset**
The board is cleared from the previous frame.
- **Place Objects**
Pacman, Ghost, and Food are placed on their current positions.
- **Show Board**
The updated grid is displayed in the console.
- **Check Collisions**
The game checks if the Ghost catches Pacman (loss condition) and if the Pacman collects Food items.
- **Movement**
The player enters a move using W, A, S, D keys, and then the Ghost moves one step randomly.

4. Ideas for the Future

The game can be improved in the following ways:

- Adding walls and obstacles to create a proper maze
- Making smart ghosts that actively chase Pacman
- Creating a graphical user interface (GUI) instead of console output
- Adding multiple levels with increasing difficulty

5. Conclusion

This project helped in understanding how to structure a Java program using OOP principles. It is a simple but functional version of Pacman and provides a strong foundation for developing more advanced games in the future.

